

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Backend platform engineer specializing in TypeScript and Node.js production systems. At Isomer AI, built and owned the full backend platform -- REST APIs, Auth0 OAuth/RBAC, Temporal.io workflow orchestration, real-time event pipelines, and multi-model AI agent tooling -- across 7 production deployments and 394 commits. Deployed containerized services on GCP Cloud Run; integrated payment and subscription workflows, document processing pipelines, and S3 file storage. Strong in event-driven architecture: AlgoPulse handles 2,000 async requests per second with Temporal and Kafka. MS CS at NC State (May 2026). Portfolio: kulkarniadiya.com

CORE COMPETENCIES

TypeScript + Node.js

REST API + Backend Platform

Auth0 + OAuth/RBAC

GCP + Cloud Run

Temporal.io + Event-Driven

Docker + Kubernetes

PostgreSQL + Redis

CI/CD + GitHub Actions

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Designed and shipped **10+ production backend features** in TypeScript, Node.js, and Express on a SaaS automation platform: REST API endpoints, Temporal.io workflow orchestration for at-least-once delivery, role-based access control via Auth0, and real-time event pipelines using Pusher -- **394 commits, 7 production deployments**, full SDLC ownership
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and a multi-model AI agent layer spanning 3 LLM providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) with MCP tooling; managed containerized deployments on GCP with Docker and Sentry observability
- Integrated subscription and billing workflow automation using Temporal durable execution; implemented PostgreSQL + Redis data layers with Prisma ORM and GitHub Actions CI/CD pipelines

NC State University Apr 2025 - Present

Graduate Research Assistant

- Deployed a cloud-native genomic data platform on Google Cloud Run (GCP) using containerized MERN stack services, serving 500GB datasets to 100+ researchers worldwide; integrated Grafana monitoring dashboards for real-time performance observability

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Automated deployments on AWS and GCP using Terraform; integrated Grafana monitoring dashboards; built 5+ AI-driven backend workflows achieving 25% accuracy improvement in automated claim processing

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed data pipelines with Kafka streaming and Hadoop + Spark batch processing, reducing ticket-processing latency by 20%; led end-to-end SDLC for a data-driven backend system serving real-time analytics

PROJECTS

AlgoPulse -- Agentic AI Platform TypeScript + Temporal + Kafka

Architected an event-driven backend platform for AI orchestration; identified API-polling bottleneck under load and re-architected to streaming Edge Functions with Temporal.io + Kafka for fault-tolerant state synchronization across distributed workers -- **2,000 async inference requests per second** without state loss. Stack mirrors Speechify's backend platform: event-driven, high-throughput, TypeScript-first.

Stack: TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka, Docker

PulseRank -- Real-Time Leaderboard and Activity Feed NestJS + REST APIs + Redis

Built a production-grade backend platform in TypeScript + NestJS: REST API layer with idempotency-key deduplication and outbox pattern, MySQL transactional writes, MongoDB feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads, and Server-Sent Events for live score streaming. Verified with Testcontainers integration tests; benchmarked with k6; shipped OpenTelemetry + Prometheus observability.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, GitHub Actions

VectorLift -- Semantic Search and Ranking Engine Python + FastAPI + Kafka

Designed and served a production search API via FastAPI with per-stage latency tracking; Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant vector store; full Docker Compose orchestration with Prometheus + Grafana observability across 6 pipeline configurations evaluated on MS MARCO.

Stack: Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker Compose

Inventory Reservation Service REST API + Concurrency

Production-grade REST API in Java 21 + Spring Boot: idempotent POST endpoints with RFC 9457 error responses, PostgreSQL row-level locking (SELECT FOR UPDATE) preventing overselling under concurrent load, automated expiry via scheduled cleanup, and multi-stage Docker build with GitHub Actions CI. Testcontainers integration tests verify exactly-one-success under simultaneous requests.

Stack: Java 21, Spring Boot 3.5, PostgreSQL, Docker, Flyway, Testcontainers, GitHub Actions

EDUCATION

North Carolina State University -- Master of Science, Computer Science Aug 2024 - May 2026

Courses: Software Engineering, Algorithms, AI, Operating Systems, Data Science, Object Oriented Development

Mumbai University (DJSCE) -- Bachelor of Technology, Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | IEEE-published: NLP document pipeline (T5 transformer, 1,500+ PDFs); Top 10 of 100+ teams HackNC24

SKILLS

- Languages:** TypeScript, JavaScript, Python, Java, SQL, C **Backend:** Node.js, NestJS, Express, FastAPI, Spring Boot, Temporal.io, Kafka, REST APIs
- Auth + Payments:** Auth0, OAuth 2.0, RBAC, JWT, subscription + billing workflow automation
- Databases:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Prisma ORM
- Cloud + Infra:** GCP (Cloud Run), AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
- Observability:** Prometheus, Grafana, Sentry, OpenTelemetry **AI/ML:** OpenAI, Anthropic, Gemini APIs, MCP, LangChain, RAG, PyTorch, FAISS